

MERN Stack Development – Module 2 Exercise

Hands-on Exercise No. 2

Course: MERN Stack Development

Module: MongoDB

Total Marks: 10

Due Date: 15/05/2026

Instructions:

- Prepare your solution in MS Word
 - Submit in .doc or .docx format
 - Late or copied submissions will not be accepted
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Learning Outcome:

- Perform CRUD operations in MongoDB
 - Understand how indexing improves performance
 - Use aggregation pipelines for data analysis
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Problem Statement:

You are assigned to design a student database for an online learning platform. The system should manage student information, enrolled courses, and quiz scores efficiently.

Task Requirements:

- Create a **students** collection with fields:
 - name
 - email
 - enrolledCourses
 - quizScores
- Insert at least 5 sample student documents
- Write queries to:
 - Find students enrolled in “MERN Stack Development”
 - Update quiz scores for one student
 - Delete a student record
- Create an index on the **email** field and demonstrate improved query performance

- Use aggregation to calculate the average quiz score per course
 - Take screenshots of all queries and their results
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Submission Guidelines:

- Upload your .docx file on LMS
- Include:
 - MongoDB queries
 - Output results
 - Screenshots of execution
- Add short explanations for each operation